MIL-R-45001
7 NOVEMBER 1957

SUPERSEDING
SAPD-P-103A
27 JANUARY 1955
SAPD-P-103
SUPPLEMENT A
4 AUGUST 1953

MILITARY SPECIFICATION

RIFLES, U. S., CAL .30, M1, M1C, AND M1D, PACKAGING OF

This specification has been approved by the Department of Defense, and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the packaging and packing of Rifle, U. S. Cal .30, M1, and Rifle, U. S. Cal .30, M1C, and M1D with equipment, for each level of protection.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

UU-T-111 — Tape; Paper, Gummed, (Sealing and Securing).

PPP-B-636 — Boxes, Fiber.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-C-843 — Cushioning Material, Cellulosic.

PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof — For Packaging and Sealing.

PPP-T-76 — Tape, Pressure — Sensitive Adhesive, Paper, Water-Resistant.

MILITARY

JAN-P-100 — Packaging and Packing for Overseas Shipment-General Specification.

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Bags, Interior Packaging.

FSC 1005

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

any pertinent data which may be of use in improving this

MIL-B-121 — Barrier Material, Greaseproofed, Flexible (Waterproofed).

MIL-B-131 — Barrier Material; Water Vaporproof, Flexible.

JAN-C-372 — Cleaner, Rifle Bore.

MIL-P-3420 — Packaging Materials, Volatile Corrosion Inhibitor Treated.

MIL-D-3464 — Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.

MIL-P-13988 — Paper, Lens, Tissue, Anti-Tarnish. Wrapping.

MIL-B-20100 — Brushes, Cleaning, for Bore and Chamber of Small Arms.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Table for Inspection of Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MS-20003 — Indicator, Humidity, Card, Three Spot Impregnated Areas.

DRAWINGS

ORDNANCE CORPS

B-7266299 — Tube, Bore, VCI Treated.

F-7267384 — Packaging and Packing for Rifle, U. S. Cal .30, M1.

(Copies of specifications, standards, drawings, publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials and methods.** The materials and methods used in preparing the Rifle, U. S. Cal .30, M1, and Rifle, U. S. Cal .30, M1C and M1D with equipment, for shipment and storage shall be in accordance with this specification and other applicable documents. The time intervals between all processes of preparation shall be of the practicable minimum.

3.1.1 *Materials.* Unless otherwise specified in the contract, the contractor shall provide the inspector certification of adherence to the requirements of material specifications for each lot of materials used. The following data shall apply wherever packaging materials are referred to in this specification by name only:

Reference	*Full identification*
Hot water	Water maintained at a minimum temperature of 200°F.
Saturated brush or patch	Saturation medium shall be rifle bore cleaner, Specification JAN-C-372.
Lens tissue	Specification MIL-P-13988.
Desiccant	Specification MIL-D-3464.
Cellulosic cushioning material	Type III, class b, thickness ⅛ inch, Specification PPP-C-843.
Pressure-sensitive tape	Type III, class 2, Specification PPP-T-60.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

Reference	*Full identification*
Grade A, greaseproof material	Type I, grade A, Specification MIL-B-121.
Moldable VCI treated material	Material conforming to type I, grade A, of Specification MIL-P-3420 (shall consist of a treated neutral kraft paper laminated to a moldable self-adhering greaseproof material) (see 6.4).

3.1.2 *Methods.* Unless otherwise specified herein, the methods and procedures for cleaning, drying, preserving, packaging, and packing shall be in accordance with the requirements of Specification MIL-P-116.

3.2 Preservation and packaging.

3.2.1 *Level A.*

3.2.1.1 *Cleaning.* The rifle shall be disassembled and the metallic surfaces of component parts and assemblies thoroughly cleaned by process C-3. Surfaces subjected to burned powder residues shall be scrubbed clean with a saturated, nonmetallic bristle brush, prior to cleaning by process C-3.

3.2.1.1.1 *Barrel assembly.* The bore and chamber shall be purged with steam or hot water, dried by procedure D-1, then followed by cleaning as specified in 3.2.1.1.1.1.

3.2.1.1.1.1 *Bore and chamber.* A minimum of three passes (forward and rearward) with a saturated close fitting wire brush, conforming to Specification MIL-B-20100, followed by three passes with individual dry patches, and one pass with a saturated patch, shall be made through the bore. The chamber shall be cleaned by the use of a saturated brush, a saturated double patch, a dry double patch and finally a saturated double patch in respective sequence. This procedure shall be repeated, as necessary to obtain the required cleanliness.

3.2.1.1.2 *Components and assemblies.* The components and assemblies removed from the rifle shall be cleaned in accordance with table I.

TABLE I. *Cleaning of components and assemblies*

Component or assembly	Process
Gas cylinder assembly	Immerse in hot water and as specified in 3.2.1.1.
Gas cylinder lock	As specified in 3.2.1.1
Gas cylinder lock screw with with valve assembly	Wire brush face of valve and clean as specified in 3.2.1.1.
Operating rod spring	As specified in 3.2.1.1
Operating rod assembly	Immerse in hot water. Wire brush piston, clean interior and exterior of rod as specified in 3.2.1.1.
Bolt assembly	As specified in 3.2.1.1
Flash hider	As specified in 3.2.1.1
Telescope (M1C and M1D) (with mount bracket)	Remove rubber eyeshield, exterior metallic surfaces cleaned with a saturated cloth.
Carrying case (M1C and M1D)	Wipe with clean dry cloth.
Cheek pad (M1C and M1D)	Wipe with clean dry cloth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

3.2.1.2 *Drying.* All moisture and residual cleaning solvents shall be thoroughly removed from the rifle prior to applying the preservative oil. Unless otherwise specified herein, all cleaned surfaces shall be dried in accordance with procedure D–1. Barrel bores and chambers shall be dried in accordance with procedure D–4 using clean dry patches. The telescope shall be dried with a clean dry cloth.

3.2.1.3 *Preservative application.*

3.2.1.3.1 The riflle shall be preserved with type P–9 oil and P–18 VCI applied as specified herein. The rifle, with the stock assembly removed, shall be coated by the dipping procedure. The metallic surfaces of the stock assembly shall be coated by the brushing procedure. The oil temperature shall be 110°F. to 120°F. The P–18 preservative application for the Rifle, U.S. Cal .30, M1, shall be in accordance with Drawing F–7267384. No preservative oil shall be applied to the cheek pad or telescope with mount bracket of Rifle, U. S. Cal .30, M1C, or M1D.

3.2.1.3.1.1 *Drainage.* The rifle shall be completely drained of excess oil prior to reassembly and packaging. The draining position shall be changed as necessary to accomplish thorough drainage.

3.2.1.4 *Packaging.*

3.2.1.4.1 *Rifle, U.S., Cal. 30, M1*

3.2.1.4.1.1 *Unit packaging.* One rifle and one sling, cushioned and wrapped as specified in 3.2.1.4.1.1.1 shall be unit packaged together in accordance with method 1A–8 as indicated on Drawing F–7267384. The VCI liner shall be folded over the butt end of the rifle and the excess air shall be expelled from the bag prior to heat sealing the closure. The bag shall be capable of passing the heat seal and quick leak tests specified in section 4.

3.2.1.4.1.1.1 *Cushioning and wrapping.* Prior to inserting the rifle and sling into the barrier bag, the front and rear sights of the rifle shall be cushioned and wrapped with moldable VCI material. The sling shall be folded flat and wrapped with type I, grade C, VCI treated material conforming to Specification MIL–P–3420. The VCI bore tube specified on the drawing shall be inserted into the barrel bore and the protruding portion bent over the muzzle end.

3.2.1.4.1.2 *Intermediate packaging.* Two rifles unit packaged in accordance with 3.2.1.4.1.1 shall be packaged in a fiberboard box in accordance with Drawing F–7267384. The barrier bag material shall be folded over the end of the stock and around the muzzle area in such a manner that the ends of the rifle can be fully inserted into the apertures of the end blocks. The rifles shall be firmly positioned in the end blocks and the box sealed with the specified tape .

3.2.1.4.2 *Rifle, U.S., Cal .30, M1C and M1D.*

3.2.1.4.2.1 *Unit packaging.* Each rifle with equipment prepackaged as specified in 3.2.1.4.2.1.. shall be unit packaged together in accordance with 3.2.1.4.2.1.1.6.

3.2.1.4.2.1.1 *Cushioning, wrapping and prepackaging.*

3.2.1.4.2.1.1.1 *Sling.* The sling shall be folded flat and wrapped with type I, grade C, VCI treated material conforming to Specification MIL–P–3420.

3.2.1.4.2.1.1.2 *Cheek pad.* The cheek pad shall be packaged in accordance with method 1C–3. The screws shall be wrapped with grade A greaseproof material prior to packaging them in the waterproof bag.

3.2.1.4.2.1.1.3 *Carrying case.* The carrying case shall be packaged in accordance with method 1C–3.

3.2.1.4.2.1.1.4 *Telescope with mount*

Source: https://assist.dla.mil -- Downloaded: 20260-22T02:11Z
Check the source to verify that this is the currentversion before use.

Rifles are unit packed in quantities of more than one, the individual

bracket. The telescope with mount bracket shall be packaged in accordance with method IIC as described herein. The lens shall be covered with at least two thicknesses of neutral lens tissue and secured to the telescope with pressure-sensitive tape. The telescope shall then be overwrapped with grade A, greaseproof material. Four units of desiccant shall be attached to the outside of the wrap with pressure-sensitive tape. A humidity indicator conforming to Standard MS 20003 shall also be attached to the outside of the wrap. The telescope with the desiccant and humidity indicator shall be overwrapped with at least three thicknesses of cellulosic cushioning material and secured with pressure-sensitive tape. The wrapped telescope shall next be placed in a water-vaporproof bag conforming to type II, class e, of Specification MIL-P-117. Excess air shall be expelled from the bag prior to heat sealing the closure.

3.2.1.4.2.1.1.4.1 The bagged telescope and the rubber eyeshield shall be packaged in a fiberboard box conforming to class 1, style RSC of Specification PPP-B-636. The box shall be sealed with 2-inch tape conforming to Specification UU-T-111.

3.2.1.4.2.1.1.5 *Rifle.* The front and rear sights of the rifle shall be cushioned and wrapped with moldable VCI material. The mount bracket on the MIC rifle shall be cusioned with moldable VCI material. A VCI bore tube conforming to Drawing B-7266299 shall be inserted into the barrel bore and the protruding portion bent over the muzzle end.

3.2.1.4.2.1.1.5.1 The rifle and one sling wrapped as specified in 3.2.1.4.2.1.1.1 shall be packaged in accordance with method 1A-8 in a barrier bag lined with VCI treated material. The bag shall be fabricated with barrier material conforming to class 1 of Specification MIL-B-131 and the VCI liner material shall conform to type I, grade C of Specification MIL-P-3420. The bag shall be so designed as to adequately inclose the rifle with sling, and to allow sufficient material after sealing to permit a subsequent heat seal (see 6.3). The VCI liner shall be folded over the butt of the rifle and the excess air removed from the bag prior to heat sealing the closure. The bag shall be capable of passing the heat seal and quick leak tests specified in section 4.

3.2.1.4.2.1.1.6 Each rifle, with equipment prepackaged as specified in 3.2.1.4.2.1.1.1 through 3.2.1.4.2.1.1.5.1, shall be unit packaged in a specially designed fiberboard box conforming to class 3, V3C of Specification PPP-B-636. The box shall be equipped with inner pads, inserts, or forms designed to adequately position, cushion, and support the rifle, with cheek pad, carrying case, and telescope in a snug-fitting manner to prevent damage to the contents during shipment and handling. The box shall be completely sealed with tape conforming to Specification PPP-T-76.

3.2.2 *Level B* (see sec. 6).

3.2.2.1 *Cleaning, drying and preservative application.* Cleaning, drying, and preservative application shall be in accordance with the requirements of 3.2.1.1, 3.2.1.2, and 3.2.1.3.

3.2.2.2 *Packaging.*

3.2.2.2.1 *Rifle, U. S., Cal .30, M1.*

3.2.2.2.1.1 *Unit package.* One rifle and one sling shall be wrapped together with moldable VCI material and unit packaged method I as described on Drawing F-7267384.

3.2.2.2.1.2 *Intermediate packaging.* Two rifles, unit packaged in accordance with 3.2.2.2.1.1 shall be packed in a fiberboard box in accordance with Drawing F-7267384. The rifles shall be firmly positioned in the end blocks and the box sealed with the specified tape.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

3.2.2.2.2 *Rifle, U. S., Cal .30, M1C and M1D.*

3.2.2.2.2.1 *Unit packaging.* One rifle and one sling shall be wrapped together with moldable VCI material. The cheek pad, carrying case, and telescope shall be prepackaged in accordance with 3.2.1.4.2.1.1.2, 3.2.1.4.2.1.1.3, and 3.2.1.4.2.1.1.4.

3.2.2.2.2.1.1 Each wrapped rifle, with the prepackaged equipment shall be unit packaged in a specially designed fiberboard box conforming to type I, class 1, of Specification PPP–B–636, with a minimum bursting strength of 275 PS1. The box shall be equipped with inner pads, inserts and forms designed to adequately position, cushion, and support the rifle, with cheek pad, carrying case, and telescope, in a snug-fitting manner so as to prevent damage to the contents during shipment and handling. The box shall be completely sealed with 3-inch sealing tape conforming to UU–T–111.

3.2.3 *Level C.*

3.2.3.1 *Cleaning, drying, preservation, and packaging.*

3.2.3.1.1 *Rifle, U. S., Cal .30, M1.* Rifles shall be cleaned, dried, preserved, and packaged in accordance with 3.2.1.1, 3.2.1.2, 3.2.1.3, 3.2.2.2.1.1, and 3.2.2.2.1.2.

3.3 Packing.

3.3.1 *Level A.*

3.3.1.1 *Rifle, U.S., Cal .30, M1.* Five intermediate packages containing two rifles each, packaged as described in 3.2.1.4.1.2, shall be packed in a nailed wood box in accordance with Drawing F–7267384.

3.3.1.2 *Rifle, U. S., Cal .30, M1C, and M1D.* Five unit packages, containing five rifles and equipment packaged as described in 3.2.1.4.2.1.1.6 shall be packed in a style 2, class 2, nailed wood box conforming to Specification PPP–B–621. Shipping containers shall be of uniform size and shall fit the contents in a manner that shall prevent movement or damage. Strapping of the container shall be in accordance with the container specification.

3.3.1.3 Completed packs prepared for shipment shall be capable of passing the rough handling test specified in Specification JAN–P–100 (see 4.5.1.)

3.3.2 *Level B* (see sec. 6).

3.3.2.1 *Rifle, U. S., Cal .30, M1.* Five intermediate packages, containing two rifles each, packaged as described in 3.2.2.2.1.2 shall be packed in a wood-cleated box in accordance with Drawing F–7267384.

3.3.2.2 *Rifle, U. S., Cal .30, M1C, and M1D.* Five unit packages containing one rifle with equipment, packaged as described in 3.2.2.2.2.1 shall be packed in a wood-cleated fiberboard box conforming to style A, (domestic type) of Specification PPP–B–591. Containers shall be of uniform size and shall fit the contents in a manner that shall prevent movement or damage. Strapping of the container shall be in accordance with the container specification.

3.3.3 *Level C.*

3.3.3.1 *Rifle, U. S., Cal .30, M1.* Ten rifles packaged as described in 3.2.3.1.1 shall be packed for shipment in accordance with Drawing F–7267384

3.4 Marking.

3.4.1 *Normal markings.* Interior packages and shipping containers shall be marked in accordance with Standard MIL–STD–129 and Drawing F–7267384 as applicable.

3.4.2 *Special markings.*

3.4.2.1 *Rifle, U. S., Cal .30 M1C, and M1D.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

Packages containing the cheek pad, carrying case, and telescope shall be clearly marked to properly identify the contents.

3.4.2.2 Level A shipping container markings shall be protected with a coating of transparent waterproof material in accordance with Standard MIL–STD–129.

3.5 **Pilot pack.** When level A or level B is specified, a pilot pack consisting of a complete and packed shipping container shall be forwarded prepaid by the contractor to the testing agency designated by the procuring activity, for approval.

3.6 **Workmanship.** Adequate controls shall be utilized to check for contamination of cleaning solvents and preservative oils. Wraps shall be conforming and firmly secured. Heat seals shall be uniform in appearance. Sealing tape shall be applied to boxes uniformly and shall adhere to joints and seams.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Inspection and tests.** The inspection procedures specified in this section are those that will be conducted by the Government. The tests shall be accomplished by the contractor or his agent under the surveillance of the Government inspector.

4.1.1 *Pilot pack.* Inspection and testing of the pilot pack (see 3.5) shall be performed at the designated testing agency. The pilot pack shall be subjected to the inspection and tests specified herein and such other inspection and tests as are necessary to determine compliance with the preservation, packaging, and packing requirements of this specification for levels A and B. The level A pilot pack shall be subjected to the rough handling test specified in 4.5.1. Acceptance of the initial lot shall be deferred pending approval of the pilot pack.

4.2 **Lot.** The lot size submitted for acceptance purposes shall be determined by the procuring activity or Government inspection agency and shall be as large as is practicable, consistent with available facilities and delivery schedules.

4.3 **Sampling.**

4.3.1 *For inspection.* Sampling for inspection shall be conducted in accordance with inspection level I of Standard MIL–STD–105.

4.3.2 *For tests.* Sampling for tests shall be conducted in accordance with inspection level L6 of appendix to Standard MIL–STD–105.

4.4 **Inspection.**

4.4.1 *Classification of defects.* Visual inspection for defects established in Table II shall apply to each item of the applicable sample of rifles, interior packages, or exterior containers as applicable.

TABLE II. *Classification of defects*

Categories and defects	AQL
Major	
1. Illegible marking	1.0
2. Incorrect marking	1.0
3. Insufficient desiccant	1.0
4. Missing parts	1.0
5. Improper level of packaging and packing	1.0
Minor	
51. Improper preservation application and drainage	2.5
52. Improper cushioning and wrapping	2.5
53. Improper closure of bags and interior packages	2.5
54. Improper blocking and bracing	2.5
55. Non-adherence to drawing requirements	2.5
56. Improper closure and strapping of shipping container	2.5
57. Rifles not properly inserted in end blocks	2.5
58. Detached parts not immobilized in rifle box	2.5
101. Workmanship	6.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

4.5 Tests.

4.5.1 *Rough handling test.* The pilot pack, for level A only, shall be subjected to the rough handling test in accordance with the requirements of Specification JAN–P–100. The drop shall be 30 inches.

4.5.2 *Heat seal and quick leak tests.* Each level A package, of the applicable test sample shall be subjected to the heat seal and quick leak tests of Specification MIL–P–116. The lot acceptance shall be based on an AQL of 6.5 percent defective for each test.

4.5.3 *Determination of cleanliness test.* From each test lot, the inspector shall subject the applicable surfaces (except for barrel bores and chambers) of each sample unit to the determination of cleanliness test in accordance with Specification MIL–P–116. Barrel bores and chambers of these units shall be tested for cleanliness by passing a clean, dry, chloride-free patch through the bore and chamber. Chemical analysis of these patches shall show no corrosion promoting residues such as chlorides. Failure of any unit to pass the above tests shall cause rejection of the lot represented, for reclean-ing.

5. PREPARATION FOR DELIVERY

(There are no applicable requirements.)

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation and packaging, and packing.

(1) Level B preservation and packaging is intended to provide adequate and economical protection during multiple domestic shipments, handling, and indoor storage periods up to 12 months from date of initial packaging.

(c) Shipping instructions for pilot pack (see 3.5).

6.2 Storage surveillance. Rifles preserved and packaged in accordance with level B requirements must be inspected to determine condition when not used within a one-year period. Rifles not used within this period should be represerved or repackaged in accordance with this specification as determined necessary by storage conditions and anticipated storage time.

6.3 Barrier bag design. Detailed information regarding barrier bags with VCI treated liners may be found on Drawing B–7265933 entitled "Bag, Barrier w/VCI Treated Liners." Copies of drawing may be obtained from the contracting officer. (See 3.2.1.4.2.1.1.5.1.)

6.4 Moldable VCI materials described in 3.1.1 are presently listed in Qualified Products List, QPL–3420–3, 2 March 1955 as 204VPI, Seal-Tite grade C, 150CA and 2300 GM.

Notice. This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restriction, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance, Department of the Army.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Ordnance
Air Force

Other interest:
Navy—MC

Preparing activity:
Army—Ordnance Corps

☆ U. S. GOVERNMENT PRINTING OFFICE: 441981—1957—(189)

Source: https://assist.dla.mil -- Downloaded: 202640-22T02:11Z
Check the source to verify that this is the currenversion before use.